Elementary Number Theory

Worked solutions

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These solutions accompany the exercises in Elementary Number Theory. The calculations retain the hypotheses used in each argument, particularly when cancellation, modular inverses, or exponent reduction is involved.

Exercise 1 — Quotients and Remainders

For each pair, we seek integers q and r satisfying $a = bq + r$ and $0 \le r < b$.
For $(a, b) = (137, 12)$, we have

$$137 = 12 \cdot 11 + 5.$$

Thus $q = 11$ and $r = 5$.
For the negative dividend, the remainder must still be nonnegative. We have

$$-137 = 12(-12) + 7.$$

Thus $q = -12$ and $r = 7$. The alternative expression $-137 = 12(-11) - 5$ does not satisfy the required remainder condition.
Finally,

$$2026 = 37 \cdot 54 + 28,$$

so $q = 54$ and $r = 28$. Therefore the three answers are

$$\boxed{(q, r) = (11, 5), \quad (-12, 7), \quad (54, 28)}.$$

Exercise 2 — Divisibility and Linear Combinations

Suppose $a \mid b$ and $a \mid c$. By definition, there exist integers s and t such that $b = as$ and $c = at$. For arbitrary $x, y \in \mathbb{Z}$,

$$bx + cy = (as)x + (at)y = a(sx + ty).$$

Since $sx + ty \in \mathbb{Z}$, the definition of divisibility gives

$$\boxed{a \mid (bx + cy)}.$$

Now suppose an integer d divides both 35 and 64. The result just proved implies that d divides every integer linear combination of these numbers. In particular,

$$11 \cdot 35 - 6 \cdot 64 = 385 - 384 = 1,$$

so $d \mid 1$. The only positive divisor of 1 is 1. Consequently, no integer greater than 1 divides both 35 and 64, and

$$\boxed{\gcd(35, 64) = 1}.$$

Exercise 3 — The Euclidean Algorithm

Applying the division algorithm repeatedly gives

$$
\begin{aligned}
12345 &= 6789 \cdot 1 + 5556,\\
6789 &= 5556 \cdot 1 + 1233,\\
5556 &= 1233 \cdot 4 + 624,\\
1233 &= 624 \cdot 1 + 609,\\
624 &= 609 \cdot 1 + 15,\\
609 &= 15 \cdot 40 + 9,\\
15 &= 9 \cdot 1 + 6,\\
9 &= 6 \cdot 1 + 3,\\
6 &= 3 \cdot 2 + 0.
\end{aligned}
$$

The last nonzero remainder is 3. Therefore,

$$\boxed{\gcd(12345, 6789) = 3}.$$

Exercise 4 — Bézout Coefficients

We substitute backward through the divisions from Exercise 3. Starting with the last nonzero remainder,

$$
\begin{aligned}
3 &= 9 - 6\\
&= 9 - (15 - 9)\\
&= 2 \cdot 9 - 15\\
&= 2(609 - 40 \cdot 15) - 15\\
&= 2 \cdot 609 - 81 \cdot 15\\
&= 2 \cdot 609 - 81(624 - 609)\\
&= 83 \cdot 609 - 81 \cdot 624\\
&= 83(1233 - 624) - 81 \cdot 624\\
&= 83 \cdot 1233 - 164 \cdot 624\\
&= 83 \cdot 1233 - 164(5556 - 4 \cdot 1233)\\
&= 739 \cdot 1233 - 164 \cdot 5556\\
&= 739(6789 - 5556) - 164 \cdot 5556\\
&= 739 \cdot 6789 - 903 \cdot 5556\\
&= 739 \cdot 6789 - 903(12345 - 6789)\\
&= -903 \cdot 12345 + 1642 \cdot 6789.
\end{aligned}
$$

Thus one pair of Bézout coefficients is

$$\boxed{x = -903, \qquad y = 1642}.$$

Direct substitution verifies the result:

$$12345(-903) + 6789(1642) = -11147535 + 11147538 = 3.$$

This agrees with $\gcd(12345, 6789) = 3$.

Exercise 5 — Euclid's Lemma

We prove the statement by induction on the number k of factors.

Base case

For $k = 1$, the hypothesis is $p \mid a_1$, which is already the desired conclusion.

Inductive step

Assume the result holds for a product of k integers. Consider integers $a_1, \ldots, a_k, a_{k+1}$ satisfying

$$p \mid a_1 a_2 \cdots a_k a_{k+1}.$$

Apply Euclid's lemma to the two integers $a_1 a_2 \cdots a_k$ and a_{k+1}. It follows that

$$p \mid a_1 a_2 \cdots a_k \quad \text{or} \quad p \mid a_{k+1}.$$

In the second case the conclusion holds immediately. In the first case, the induction hypothesis implies $p \mid a_i$ for some $i \in \{1, \ldots, k\}$. Thus in either case p divides at least one of the $k + 1$ factors.

By induction,

$$\boxed{p \mid a_1 a_2 \cdots a_k \implies \exists i \in \{1, \ldots, k\} \text{ such that } p \mid a_i}.$$

□

Exercise 6 — Congruence and Remainders

Let the division algorithm give

$$a = nq + r, \qquad b = nq' + r', \qquad 0 \le r, r' < n.$$

Suppose first that $a \equiv b \pmod{n}$. Then $n \mid (a - b)$. Since

$$a - b = n(q - q') + (r - r'),$$

it follows that $n \mid (r - r')$. However, $-(n - 1) \le r - r' \le n - 1$, and the only multiple of n in this interval is 0. Hence $r = r'$, so a and b have the same remainder.

Conversely, suppose $r = r'$. Then

$$a - b = n(q - q'),$$

so $n \mid (a - b)$. By definition, $a \equiv b \pmod{n}$. Therefore,

$$\boxed{a \equiv b \pmod{n} \iff a \text{ and } b \text{ have the same remainder modulo } n}.$$

□

Exercise 7 — Modular Arithmetic

First reduce the inputs modulo 17:

$$38 \equiv 4 \pmod{17}, \qquad 49 \equiv -2 \pmod{17}.$$

For the sum,

$$38 + 49 \equiv 4 + (-2) \equiv 2 \pmod{17}.$$

For the product,

$$38 \cdot 49 \equiv 4(-2) \equiv -8 \equiv 9 \pmod{17}.$$

For the power,

$$38^4 \equiv 4^4 = 256 = 17 \cdot 15 + 1 \equiv 1 \pmod{17}.$$

The requested canonical representatives are therefore

$$\boxed{2, \qquad 9, \qquad 1}.$$

Exercise 8 — Modular Inverses

An inverse of a modulo n exists if and only if $\gcd(a, n) = 1$.

The inverse of 17 modulo 3120

The Euclidean algorithm gives

$$\begin{aligned} 3120 &= 17 \cdot 183 + 9, \\ 17 &= 9 \cdot 1 + 8, \\ 9 &= 8 \cdot 1 + 1. \end{aligned}$$

Substituting backward,

$$\begin{aligned} 1 &= 9 - 8 \\ &= 9 - (17 - 9) \\ &= 2 \cdot 9 - 17 \\ &= 2(3120 - 17 \cdot 183) - 17 \\ &= 2 \cdot 3120 - 367 \cdot 17. \end{aligned}$$

Thus -367 is an inverse of 17 modulo 3120. Its canonical representative is $3120 - 367 = 2753$, so

$$\boxed{17^{-1} \equiv 2753 \pmod{3120}}.$$

Indeed, $17 \cdot 2753 = 15 \cdot 3120 + 1$.

The inverse of 18 modulo 3120

Since $\gcd(18, 3120) = 6 \neq 1$, the invertibility criterion gives

$$\boxed{18^{-1} \pmod{3120} \text{ does not exist}}.$$

The inverse of 35 modulo 64

The Euclidean algorithm and backward substitution give

$$\begin{aligned} 64 &= 35 + 29, \\ 35 &= 29 + 6, \\ 29 &= 4 \cdot 6 + 5, \\ 6 &= 5 + 1, \end{aligned}$$

and hence

$$\begin{aligned} 1 &= 6 - 5 \\ &= 6 - (29 - 4 \cdot 6) \\ &= 5 \cdot 6 - 29 \\ &= 5(35 - 29) - 29 \\ &= 5 \cdot 35 - 6(64 - 35) \\ &= 11 \cdot 35 - 6 \cdot 64. \end{aligned}$$

Therefore,

$$\boxed{35^{-1} \equiv 11 \pmod{64}}.$$

Exercise 9 — Linear Congruences

The congruence $9x \equiv 12 \pmod{15}$

We have $\gcd(9, 15) = 3$, and $3 \mid 12$, so the congruence has exactly three solutions modulo 15. Dividing the equation and modulus by 3 gives

$$3x \equiv 4 \pmod{5}.$$

The inverse of 3 modulo 5 is 2, so

$$x \equiv 8 \equiv 3 \pmod{5}.$$

The three lifts modulo 15 are

$$\boxed{x \equiv 3, 8, 13 \pmod{15}}.$$

The congruence $14x \equiv 5 \pmod{21}$

Here $\gcd(14, 21) = 7$, but $7 \nmid 5$. The solvability criterion therefore implies that

$$\boxed{14x \equiv 5 \pmod{21} \text{ has no solution}}.$$

Directly, a solution would give $14x - 5 = 21k$ for some $k \in \mathbb{Z}$, or $5 = 7(2x - 3k)$, which would imply $7 \mid 5$, a contradiction.

Exercise 10 — Units and the Totient Function

An element $[a]_{20}$ is a unit precisely when $\gcd(a, 20) = 1$. Testing the canonical representatives gives

$$\mathbb{Z}_{20}^{*} = \{[1]_{20}, [3]_{20}, [7]_{20}, [9]_{20}, [11]_{20}, [13]_{20}, [17]_{20}, [19]_{20}\}.$$

There are eight elements, so

$$\boxed{\varphi(20) = 8}.$$

Since $20 = 2^2 \cdot 5$ and 4 and 5 are coprime, multiplicativity and the prime-power formula give

$$\varphi(20) = \varphi(2^2)\varphi(5) = (4-2)(5-1) = 2 \cdot 4 = 8.$$

We may also count directly in $\{1, \ldots, 20\}$. There are 10 multiples of 2 and 4 multiples of 5. The 2 multiples of 10 appear in both groups, so inclusion-exclusion gives

$$10 + 4 - 2 = 12$$

integers that are not coprime to 20. The remaining $20 - 12 = 8$ integers are coprime to 20, confirming the result.

Exercise 11 — Cancellation

The congruence $ac \equiv bc \pmod{n}$ means

$$n \mid c(a-b).$$

Since $\gcd(c, n) = 1$, Bézout's identity provides integers u, v such that

$$cu + nv = 1.$$

Multiplying by $a - b$ gives

$$a - b = cu(a-b) + nv(a-b).$$

The first term on the right is divisible by n because $n \mid c(a-b)$, and the second is visibly divisible by n. Hence $n \mid (a-b)$, which proves

$$\boxed{a \equiv b \pmod{n}}.$$

The coprimality hypothesis is necessary. For example, take $a = 1$, $b = 4$, $c = 2$, and $n = 6$. Then

$$ac = 2 \equiv 8 = bc \pmod{6},$$

but $1 \not\equiv 4 \pmod{6}$. Here $\gcd(2, 6) = 2 \neq 1$.

Exercise 12 — Euler's Theorem

Since $\gcd(7, 20) = 1$, Euler's theorem applies. We have $\varphi(20) = 8$, so

$$7^8 \equiv 1 \pmod{20}.$$

Because $222 = 8 \cdot 27 + 6$,

$$7^{222} \equiv 7^6 \pmod{20}.$$

Now $7^2 = 49 \equiv 9 \pmod{20}$ and $7^4 \equiv 9^2 = 81 \equiv 1 \pmod{20}$. Therefore,

$$7^6 = 7^4 \cdot 7^2 \equiv 1 \cdot 9 \equiv 9 \pmod{20},$$

and the answer is

$$\boxed{7^{222} \equiv 9 \pmod{20}}.$$

Checking $\gcd(7, 20) = 1$ before reducing the exponent is essential because Euler's theorem does not assert $a^{\varphi(n)} \equiv 1 \pmod{n}$ for nonunits a.

Exercise 13 — Fast Modular Exponentiation

The binary decomposition of the exponent is

$$117 = 64 + 32 + 16 + 4 + 1.$$

Starting with 5 and repeatedly squaring modulo 19 gives

$$\begin{aligned}
5^1 &\equiv 5 \pmod{19},\\
5^2 &\equiv 25 \equiv 6 \pmod{19},\\
5^4 &\equiv 6^2 \equiv 36 \equiv 17 \pmod{19},\\
5^8 &\equiv 17^2 \equiv 289 \equiv 4 \pmod{19},\\
5^{16} &\equiv 4^2 \equiv 16 \pmod{19},\\
5^{32} &\equiv 16^2 \equiv 256 \equiv 9 \pmod{19},\\
5^{64} &\equiv 9^2 \equiv 81 \equiv 5 \pmod{19}.
\end{aligned}$$

Using the powers selected by the binary expansion,

$$\begin{aligned}
5^{117} &= 5^{64}5^{32}5^{16}5^{4}5\\
&\equiv 5 \cdot 9 \cdot 16 \cdot 17 \cdot 5\\
&\equiv 1 \pmod{19}.
\end{aligned}$$

Therefore,

$$\boxed{5^{117} \equiv 1 \pmod{19}}.$$

Exercise 14 — The Chinese Remainder Theorem

We first combine the congruences modulo 3 and 5. The integers congruent to 3 modulo 5 begin

$$3, 8, 13, 18, \ldots,$$

and $8 \equiv 2 \pmod{3}$. Thus

$$x \equiv 8 \pmod{15}.$$

Write $x = 8 + 15k$. The remaining congruence requires

$$8 + 15k \equiv 2 \pmod{7}.$$

Since $8 \equiv 1 \pmod{7}$ and $15 \equiv 1 \pmod{7}$, this becomes

$$1 + k \equiv 2 \pmod{7},$$

so $k \equiv 1 \pmod{7}$. Taking $k = 1$ gives $x = 23$. Because 3, 5, and 7 are pairwise coprime, the Chinese remainder theorem guarantees uniqueness modulo $3 \cdot 5 \cdot 7 = 105$. Hence

$$\boxed{x \equiv 23 \pmod{105}}.$$

Verification gives

$$23 \equiv 2 \pmod 3, \qquad 23 \equiv 3 \pmod 5, \qquad 23 \equiv 2 \pmod 7.$$

Exercise 15 — Toy RSA

With $p = 11$ and $q = 17$, we obtain

$$n = pq = 187$$

and

$$\varphi(n) = (p-1)(q-1) = 10 \cdot 16 = 160.$$

Since $\gcd(7, 160) = 1$, the exponent $e = 7$ is invertible modulo 160. In fact,

$$7 \cdot 23 = 161 \equiv 1 \pmod{160},$$

so

$$\boxed{d = 23}.$$

Encryption

Repeated squaring modulo 187 gives

$$8^2 \equiv 64 \pmod{187}, \qquad 8^4 \equiv 64^2 \equiv 169 \pmod{187}.$$

Since $7 = 4 + 2 + 1$,

$$8^7 \equiv 169 \cdot 64 \cdot 8 \equiv 134 \pmod{187}.$$

Thus the ciphertext representative is

$$\boxed{c = 134}.$$

Decryption

We compute powers of 134 modulo 187:

$$\begin{aligned} 134^2 &\equiv 4 \pmod{187}, \\ 134^4 &\equiv 16 \pmod{187}, \\ 134^8 &\equiv 69 \pmod{187}, \\ 134^{16} &\equiv 86 \pmod{187}. \end{aligned}$$

Because $23 = 16 + 4 + 2 + 1$, decryption yields

$$\begin{aligned} 134^{23} &= 134^{16}134^4134^2134 \\ &\equiv 86 \cdot 16 \cdot 4 \cdot 134 \\ &\equiv 8 \pmod{187}. \end{aligned}$$

Thus decryption recovers the original representative $m = 8$.

Exercise 16 — Degenerate Linear Congruences

For $0x \equiv 0 \pmod 7$, every integer x is a solution. There are exactly seven solution classes:

$$\boxed{[0]_7, [1]_7, [2]_7, [3]_7, [4]_7, [5]_7, [6]_7}.$$

Here $d = \gcd(0, 7) = 7$, and $d \mid 0$, so the count agrees with the general criterion.

For $0x \equiv 3 \pmod 7$, the left side is always congruent to zero, whereas $7 \nmid 3$. There are no solutions, again in agreement with $d = \gcd(0, 7) = 7$.

Finally, $14x \equiv 21 \pmod 7$ reduces to $0 \equiv 0 \pmod 7$ for every integer x. Since $\gcd(14, 7) = 7$ divides 21, all seven residue classes are solutions.

In each case, the coefficient of x is divisible by the modulus. It is enough to check whether the right-hand side is also divisible by the modulus. No inversion, and in particular no inverse modulo 1, is needed.

Exercise 17 — RSA for a Nonunit Message

Since $\gcd(5, 55) = 5$, the message $m = 5$ is not a unit modulo 55. Euler's theorem therefore does not justify reducing powers of 5 modulo $\varphi(55) = 40$.

Encryption nevertheless gives

$$c \equiv 5^3 = 125 \equiv 15 \pmod{55}.$$

For decryption, first work modulo 5. Since $15 \equiv 0 \pmod 5$ and $27 > 0$, we have $15^{27} \equiv 0 \pmod 5$.

Modulo 11, we have $15 \equiv 4$ and $\gcd(4, 11) = 1$. Fermat's little theorem permits reduction of the exponent modulo 10, so

$$\begin{aligned} 15^{27} &\equiv 4^7 \pmod{11}, \\ 4^2 &\equiv 5 \pmod{11}, \\ 4^4 &\equiv 3 \pmod{11}, \\ 4^7 = 4^4 \cdot 4^2 \cdot 4 &\equiv 3 \cdot 5 \cdot 4 \equiv 5 \pmod{11}. \end{aligned}$$

The recovered representative must consequently satisfy

$$m' \equiv 0 \pmod 5, \qquad m' \equiv 5 \pmod{11}.$$

The integer 5 satisfies both congruences. By the Chinese remainder theorem it is the unique solution in $\{0, \dots, 54\}$, so

$$\boxed{15^{27} \equiv 5 \pmod{55}}.$$

The correctness proof treats the prime divisor 5 through the case where the message is zero modulo that prime. It treats the prime divisor 11 through Fermat's little theorem. These two cases explain why RSA arithmetic recovers this message even though it is not a unit modulo the composite modulus.